

1. A bank wants to predict whether a loan applicant will default based on credit score, income, and past loan history. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type:

step 1: Machine Learning

step 2: supervised learning

step 3: classification problem (loan applicant -> default or not default)

Step-By-Step Logic:

1. Load the loan_applicant dataset
(Customer_ID, Age, Gender, Marital_Status, Dependents, Employment_Type, Work_Experience_Years, Annual_Income, Monthly_Income, Existing_Loan_Amount, Credit_Card_Outstanding, Debt_To_Income_Ratio, Credit_Score, Number_Of_Past_Loans, Past_Loan_Defaults, Payment_History_Score, Credit_Utilization_Ratio, Loan_Amount, Loan_Term_Months, Loan_Purpose, Interest_Rate, Loan_Default)
 2. Convert the categorical column into numerical column using `pd.get_dummies` (Gender, Marital_Status, Employment_Type, Loan_Purpose)
 3. Split input (Customer_ID, Age) and output data (Loan_Default)
 4. Split test and train data
 5. Data preprocessing using standardization (if required)
 6. Model creation (using algorithms such as logistic regression, KNN, NB, SVC, decision tree, random forest ...)
 7. Test the model
 8. Model evaluation using confusion_matrix
 9. Finalize the best model
2. A retail store wants to predict the demand for different products to optimize inventory levels. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type:

step 1: Machine Learning

step 2: unsupervised learning

step 3: Regression Problem (product_sold_units – continuous numbers)

(To optimize inventory levels, the problem type depends on the objective. If we predict the number of units sold, it is a regression problem. If we categorize demand into levels such as high, medium, or low, it becomes a classification problem. If no predefined target is available and we aim to group products based on similar demand patterns, then it is a clustering problem. The clustered results can then be analyzed and prioritized for inventory planning and visualization.)

Step-By-Step Logic:

1. Load the retail store dataset
(Date,Store_ID,Store_Type,Location,Product_ID,Product_Category,Brand,MRP,Units_Sold,Inventory_Level,Reorder_Point,Discount_Percentage,Promotion,Holiday,Competitor_Price,Weather,Fuel_Price,CPI, units_sold)
2. Convert the categorical column into numerical column
3. Split input (Date,Store_ID,Store_Type..) and output data (units_sold - product sold units)
4. Split train and test data
5. Data preprocessing using standardization (if required)
6. Model creation using regression algorithms (linear regression, SVR, decision tree regressor, random forest, boosting algorithms etc..)
7. Test the model (y_prediction)
8. Model evaluation using r2_score
9. Finalize the best regression models based on the results of all models

3. A factory wants to detect whether a manufactured product is defective based on sensor readings and quality control data. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type:

step 1: Machine Learning

step 2: supervised learning

step 3: classification problem (classify manufactured product – defective or not defective)

Step-By-Step Logic:

1. Load the manufacturing product dataset
(Product_ID,Temperature_C,Pressure_PSI,Humidity_%,Vibration_Level,Voltage_V,Current_A,Production_Speed,Machine_Runtime_Hours,Operator_Experience_Years,Raw_Material_Quality_Score,Maintenance_Last_Days,Defective)
2. Convert the categorical column into numerical column using pd.get_dummies
3. Split input (Product_ID,Temperature_C ...) and output data (Defective)
4. Split test and train data
5. Data preprocessing using standardization (if required)
6. Model creation (using algorithms such as logistic regression,KNN, NB, SVC, decision tree, random forest ...)
7. Test the model

8. Model evaluation using confusion_matrix
9. Finalize the best model

4. A healthcare provider wants to analyze patient symptoms and classify them into different disease categories. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type:

step 1: Machine Learning

step 2: supervised learning

step 3: classification problem (multi classification problem : classify – disease category)

Step-By-Step Logic:

1. Load the patient symptoms dataset
(Patient_ID, Age, Gender, BMI, Fever, Cough, Fatigue, Headache, Shortness_of_Breath, Chest_Pain, Nausea, Joint_Pain, Sore_Throat, Runny_Nose, WBC_Count, Blood_Pressure, Blood_Sugar, Disease_Category)
2. Convert the categorical column into numerical column using `pd.get_dummies`
3. Split input (Patient_ID, Age, Gender, BMI, Fever, Cough) and output data (Disease_Category)
4. Split test and train data
5. Data preprocessing using standardization (if required)
6. Model creation (using algorithms such as logistic regression, KNN, NB, SVC, decision tree, random forest ...)
7. Test the model
8. Model evaluation using confusion_matrix
9. Finalize the best model

5. An e-commerce company wants to identify and remove fake reviews posted by bots or fraudsters. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type:

step 1: Machine Learning

step 2: unsupervised learning

step 3: classification problem (classify – fake review or genuine review)

Step-By-Step Logic:

1. Load the e-commerce review dataset
(Review_ID, User_ID, Account_Age_Days, Total_Reviews_By_User, Verified_Purchase, User_Average_Rating, Reviewer_IP_Flag, Product_

ID,Product_Category,Product_Average_Rating,Review_Text,Review_Length,Sentiment_Score,Excessive_Capital_Words,Repeated_Words_Count,Review_Posting_Gap,Rating,Rating_Deviation,Fake_Review)

2. Convert the categorical column into numerical column using `pd.get_dummies (Product_Category , Review_Text)`
3. Split input (Review_ID,User_ID,Account_Age_Days ..) and output data (Fake_Review)
4. Split test and train data
5. Data preprocessing using standardization (if required)
6. Model creation (using algorithms such as logistic regression,KNN, NB, SVC, decision tree, random forest ...)
7. Test the model
8. Model evaluation using `confusion_matrix`
9. Finalize the best model

6. A financial firm wants to predict stock price movements based on historical price data and market indicators. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type:

step 1: Machine Learning

step 2: supervised learning

step 3: classification problem (stock price -> increase or decrease)

Step-By-Step Logic:

1. Load the stock_price dataset
(Date,Stock_Symbol,Sector,Open_Price,High_Price,Low_Price,Close_Price,Adjusted_Close,Volume,SMA_10,SMA_50,EMA_10,RSI,MACD,Bollinger_Upper,Bollinger_Lower,Volatility,Market_Index_Close,VIX,Interest_Rate,Inflation_Rate, Price_Movement)
2. Convert the categorical column into numerical column using `pd.get_dummies (Stock_Symbol,Sector)`
3. Split input (Date,Stock_Symbol) and output data (Price_Movement)
4. Split test and train data
5. Data preprocessing using standardization (if required)
6. Model creation (using algorithms such as logistic regression,KNN, NB, SVC, decision tree, random forest ...)
7. Test the model
8. Model evaluation using `confusion_matrix`
9. Finalize the best model

7. A social media platform wants to detect fake user accounts based on user activity and profile data. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type:

step 1: Machine Learning

step 2: supervised learning

step 3: classification problem (predict – fake user account or genuine user account)

Step-By-Step Logic:

1. Load the social media platform dataset
(User_ID,Account_Age_Days,Profile_Picture,Bio_Length,Username_Length,Has_Numbers_In_Username,Followers_Count,Following_Count,Followers_Following_Ratio,Total_Posts,Posts_Per_Day,Likes_Per_Post,Comments_Per_Post,DM_Sent_Per_Day,Reported_Count,Login_Frequency_Per_Day,Fake_Account)
2. Split input (User_ID,Account_Age_Days,Profile_Picture ...) and output data (Fake_Account)
3. Split test and train data
4. Data preprocessing using standardization (if required)
5. Model creation (using algorithms such as logistic regression,KNN, NB, SVC, decision tree, random forest ...)
6. Test the model
7. Model evaluation using confusion_matrix
8. Finalize the best model

8. A marketing agency wants to segment customers into different groups based on their purchasing behavior. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type:

step 1: Machine Learning

step 2: unsupervised learning

step 3: clustering

Step-By-Step Logic:

1. Load the marketing agency – customer dataset
(Customer_ID,Age,Gender,Annual_Income,Total_Purchases,Avg_Purchase_Value,Total_Spending,Purchase_Frequency_Per_Month,Recency_Days,Online_Purchase_Percentage,Discount_Usage_Percentage>Returns_Count)

2. Select the important features for the X - input data
(Annual_Income, Total_Purchases, Avg_Purchase_Value, Total_Spending, Purchase_Frequency_Per_Month, Recency_Days, Online_Purchase_Percentage, Discount_Usage_Percentage, Returns_Count)
3. Find the number of clusters (based on the methods – if required for the clustering algorithm(k means, etc..))
4. Data preprocessing using standardization (if required)
5. Model creation & prediction of y_labels – using clustering algorithms (k means, agglomerative, DBSCAN, HDBSCAN, optics etc..)
6. Create new dataset with cluster_group (by prioritizing the product demand) and Save it as csv file
7. Visualize the clusters using seaborn.Implot – (for visual inspection of clusters)
8. Cluster model evaluation using cluster evaluation metrics
9. Finalize the best clustering models based on the results and use case requirements – (by analyzing the cluster outputs, evaluation metric, domain knowledge and visual cluster plot)

9. A geospatial research team wants to analyze satellite images to classify different land types (forest, water, urban). What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type:

step 1: Machine Learning

step 2: supervised learning

step 3: classification problem (classify land type – forest,water,urban)

Step-By-Step Logic:

1. Load the satellite images dataset
(Image_ID, Latitude, Longitude, Resolution_m, Avg_Red, Avg_Green, Avg_Blue, NDVI, Texture_Contrast, Edge_Density, Water_Index, Land_Type)
2. Convert the categorical column into numerical column using pd.get_dummies (Land_Type)
3. Split input (Image_ID, Latitude, Longitude ...) and output data (Land_Type)
4. Split test and train data
5. Data preprocessing using standardization (if required)
6. Model creation (using algorithms such as logistic regression, KNN, NB, SVC, decision tree, random forest ...)
7. Test the model
8. Model evaluation using confusion_matrix
9. Finalize the best model

10. A streaming service wants to predict which users are likely to cancel their subscriptions. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type:

step 1: Machine Learning

step 2: unsupervised learning

step 3: classification problem (classify – user will continue subscription or user will cancel the subscription)

Step-By-Step Logic:

1. Load the streaming service users dataset (User_ID, Age, Gender, Country, Subscription_Type, Subscription_Length_Months, Avg_Screen_Time_Min, Avg_Watch_Hours_Per_Week, Num_Devices, Num_Shows_Watched, Num_Movies_Watched, Last_Login_Days_Ago, Support_Tickets, Payment_Delay_Count, Churn)
2. Convert the categorical column into numerical column using `pd.get_dummies (Gender, Country, Subscription_Type)`
3. Split input (User_ID, Age, Gender, Country, Subscription type ...) and output data (Churn)
4. Split test and train data
5. Data preprocessing using standardization (if required)
6. Model creation (using algorithms such as logistic regression, KNN, NB, SVC, decision tree, random forest ...)
7. Test the model
8. Model evaluation using confusion_matrix
9. Finalize the best model